

Warm-up 1

a) Compute the characteristic polynomial of the following generic 2×2 matrix:

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

b) What are the determinant and the trace of the matrix in part (a)?

Now compute the eigenvalues of the above matrix using the quadratic formula.

During class 1

Which of the following vectors is largest in the usual (Euclidean, L^2) norm?

$$\begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

Which of the following vectors is largest in the taxicab (L^1) norm?

$$\begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

Which of the following vectors is largest in the max (L^∞) norm?

$$\begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$